Q2.1

By examining the spectrogram before 2 seconds, several horizontal frequency bands can be observed, roughly around 500–600 Hz and around 1000 Hz. These bands correspond to the fundamental frequencies and their harmonics. However, due to overlapping harmonics and limited resolution, it is difficult to precisely determine the notes using the spectrogram alone.

0s: 500 HZ 1060HZ(overtone)

0.25s: 600HZ

0.5s: 1000Hz

0.78s: 500Hz

1.1s: 600Hz

1.35s 800Hz

Therefore, the Fast Fourier Transform (FFT) was applied to short segments between 2 and 3 seconds to obtain more accurate frequency information.

2s: 589.66Hz and 1179Hz (overtone of 589Hz)

At 2.0 seconds, a dominant peak is observed at approximately 589.66 Hz, which is close to D5 (587.33 Hz). A second peak at around 1179 Hz corresponds to the second harmonic of the fundamental frequency.

2.2s: 499Hz, 589.66Hz is decaying

At 2.2 seconds, peaks at approximately 499 Hz and 589 Hz are observed, indicating the presence of B4 (493.88 Hz) and D5, with D5 gradually decaying.

2.4s: 589Hz, 499Hz is decaying

2.6s: 589Hz

At 2.4 seconds and 2.6 seconds, the dominant frequency remains around 589 Hz, confirming the persistence of D5.

2.8s: 589Hz 1059Hz

3s: 499Hz 589Hz 999Hz 1059Hz

At 2.8 seconds and 3.0 seconds, multiple peaks appear (approximately 499 Hz, 589 Hz, and around 1000–1059 Hz), suggesting overlapping notes and harmonics.

Reasons for parameter choices:

I selected the first 5 seconds of the audio because this portion is less dense and easier to analyze. When fewer notes overlap, it becomes easier to identify individual pitches.

Window length (4096):
I used a relatively large window size to improve frequency resolution. Since identifying piano notes relies on accurately detecting frequencies, a larger window makes the spectral peaks clearer and easier to interpret.

Overlap (3072):
A large overlap was chosen to ensure smooth transitions between frames. We only move forward a bit a time.

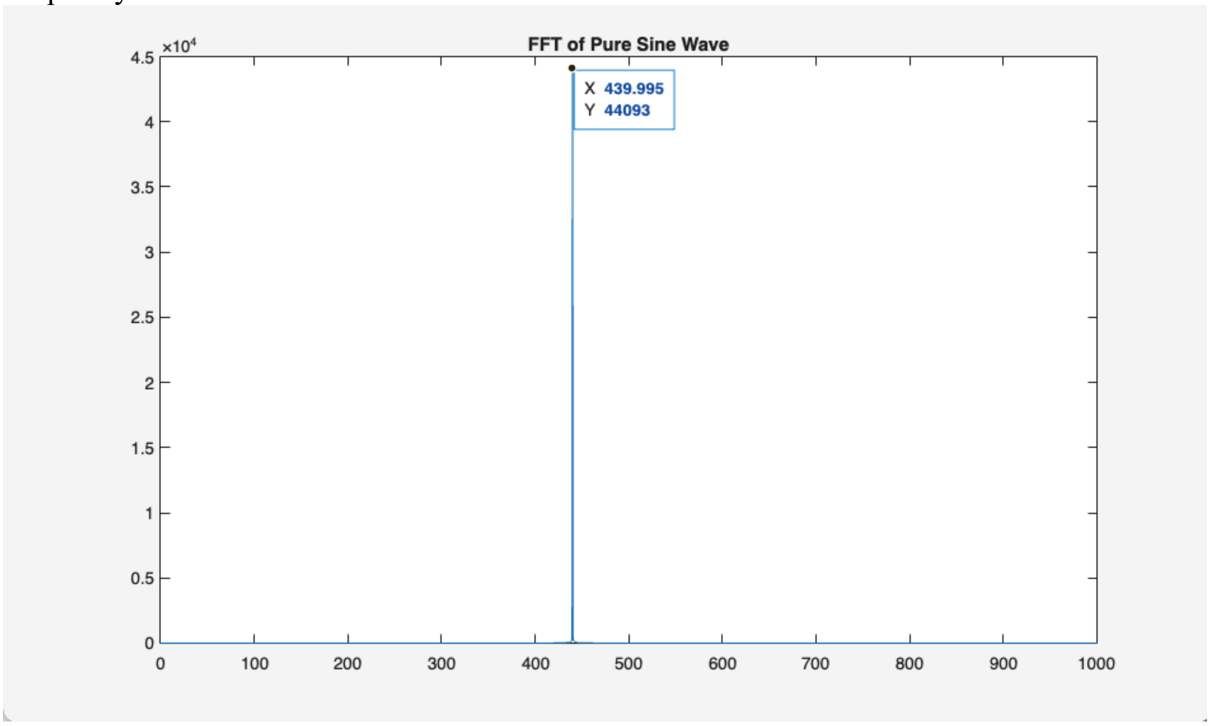
NFFT (number of fast fourier transform) (8192):
I used an NFFT larger than the window length to create a denser frequency grid. The unit frequency gap is ~5Hz so that we can differentiate among 440Hz,445Hz, 450Hz etc.

Magnitude scaling (dB):
The magnitude was converted to decibels using 20*log10(abs(S) + 1e-6). The dB scale allows both strong and weak frequency components to be visible, which is important for observing harmonics in piano sounds.

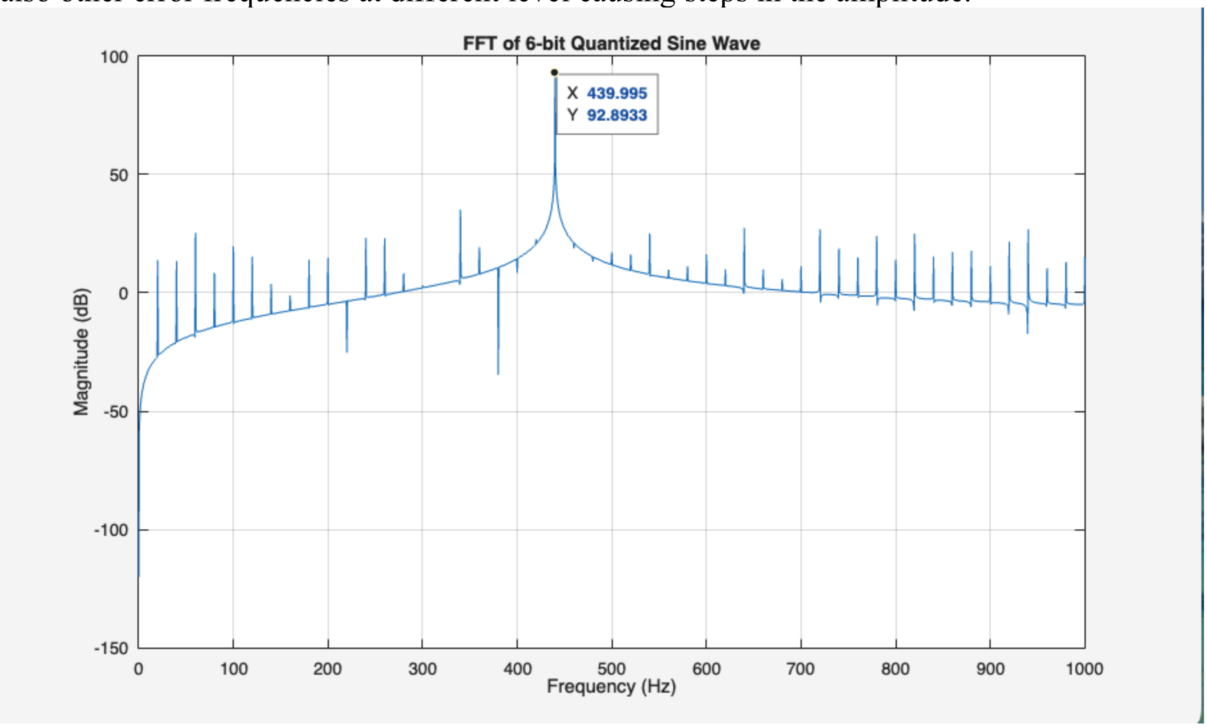
Frequency range (0–2000 Hz):
The frequency axis was limited to 0–2000 Hz because most of the relevant information for identifying piano notes lies in the fundamental frequencies and lower harmonics.

Q2. 2

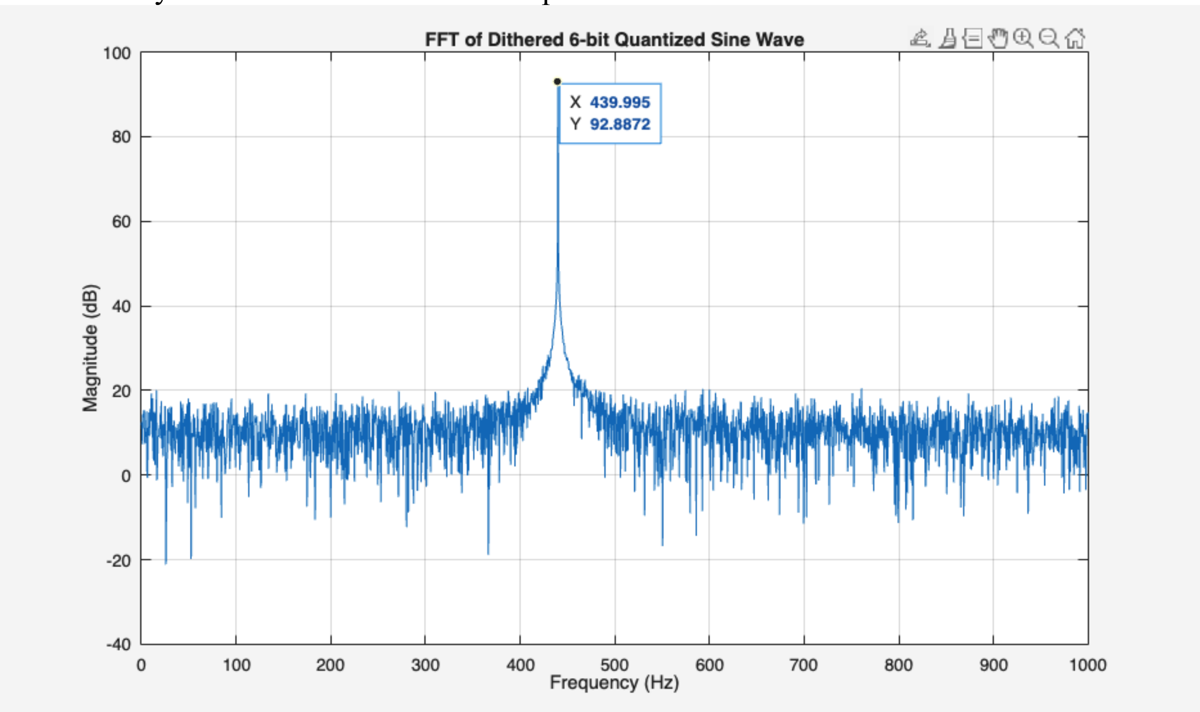
In the spectrum of the original signal: the graph is smooth, with only one single spike at frequency = 440Hz.



In the spectrum of quantized signal: there is still the biggest spike at frequency = 440Hz but also other error frequencies at different level causing steps in the amplitude.



In the spectrum of dithered signal: The error is spread out randomly, making the error less noticeable by human ear. We can see the spectrum is smoother too.



In the undithered signal, the spectrum shows extra sharp peaks caused by quantization distortion. These peaks are structured and sound harsh.
In the dithered signal, the distortion is spread out as noise instead of sharp peaks. This makes it sound smoother and less unpleasant, since random noise is less noticeable than structured distortion.

Q3:

I generated a blurred and noisy version of the cameraman image from what is described in the lecture. From the original formula of Wiener Filter, I am able to implement it to degraded images.

The degraded image was restored using two different power spectra: the true cameraman image spectrum and the spectrum of a building image. Multiple reconstructions were generated by varying the noise power parameter, and the best results were selected based on visual quality and the sum of squared errors (SSE).I generated ~10 images for both experiments and found that when parameters are finetuned to the best, using building power spectrum is worse than using cameraman spectrum.

The resulting SSE values are:

- Cameraman spectrum: 201
- Building spectrum: 260

The result using the cameraman spectrum produced a lower SSE and visually better reconstruction. This is because the Wiener filter relies on an accurate estimate of the image power spectrum. Using an incorrect spectrum (building) leads to poorer restoration.